README

ApexPlanet Data Analytics Internship — Task 4

Project title

Data Storytelling & Statistical Validation of Sales Data

Project overview

This project analyzes the supplied sales dataset to identify category-wise, city-wise, and monthly sales patterns. It also applies one-way ANOVA to test whether mean transaction sales differ across product categories.

Files in this submission

File	Purpose
Task4_Final_Presentation.pdf	Visual summary of the story, findings, statistical test, and recommendations.
Hypothesis_Testing_Summary.pdf	Hypotheses, ANOVA results, category summary, decision, and limitations.
Data_Storytelling_Report.pdf	Written business narrative, findings, interpretation, and recommendations.
Hypothesis_Testing_Code.pdf	Readable Python code for running the ANOVA test.
README.pdf	Project overview and instructions.

Tools and libraries

Python, pandas, SciPy, openpyxl, and ReportLab. The analysis uses Total_Sales as the numeric outcome and Category as the grouping variable.

How to reproduce the analysis

1. Place the Excel dataset and the Python code in the same folder.
2. Install dependencies: pip install pandas scipy openpyxl
3. Run the code in Hypothesis_Testing_Code.pdf by copying it into a .py file or Jupyter Notebook.
4. Confirm the printed category summaries, F-statistic, p-value, and decision.
5. Compare results with the Hypothesis Testing Summary and written report.

Main result

Total recorded sales: ■139,399,439.65. Top category by revenue: Electronics (■50,778,581.70). Top city by revenue: Patna (■19,285,966.89). ANOVA p-value: 0.580286. At $\alpha = 0.05$, the decision is to fail to reject H■.

Limitations

This is a descriptive analysis of observational data, not a causal experiment. Revenue is not the same as profit. Statistical conclusions depend on ANOVA assumptions; review data quality and assumptions before using results for important business decisions.

Prepared for the ApexPlanet Data Analytics Internship, Task 4.